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Configuration interaction as an Anderson impurity solver

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Abstract

We present an implementation of an Anderson impurity solver using a configuration interaction approach implemented inside the package FERMIONS. By combining a bit and vector representation we can improve the calculation of a power iteration for a given Hamiltonian and starting state by up to a factor of 2000 compared to the existing bit only representation. This will allow us to calculate systems with an impurity coupled to hundreds of bath sites in the future.

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1 Introduction

Many body physics can give rise to interesting phenomena, e.g., Bose-Einstein condensates (BEC) or superconductivity. The general problem however is the exponential increase of configurations with each particle added. One approach to overcome this is dynamical mean-field theory (DMFT) [1], an approximation that is controlled in $1/z$ (z : number of neighbors) and which successfully describes Kondo physics [2, 3] and Mott insulators [4]. In DMFT, we can map a crystal lattice onto an Anderson impurity model (AIM) [5], which we can solve among other things with quantum Monte Carlo (QMC) [6, 7] or exact diagonalization (ED) [8] methods. The disadvantages of the former compared to the latter are that QMC is computationally demanding for low temperatures and the analytical continuation from imaginary frequencies is an ill-conditioned problem [9]. The problem with ED is that it is only feasible for a few bath sites which works well for isolators but not for materials with a Kondo peak. Since we are interested in the ground state of a system and states close to it, we can limit the number of excitations, using configuration interaction (CI) [10, 11].

The goal of this thesis is building a framework which can reproduce the work of Lu et al. [12].

1.1 Anderson impurity model

Since no analytical solution of the Hubbard model [13] in arbitrary dimensions exists, we map it within DMFT onto a single-site AIM to describe our system with fewer degrees of freedom. In DMFT we hence ignore all spatial fluctuations and only keep time correlations. This approximation becomes exact in the limit of infinite lattice coordination [14]. Our system contains a single impurity i with energy level ϵ and on-site Coulomb interaction U . This impurity is coupled by a hybridization V_k to bath sites with energies ϵ_k as illustrated in Fig. 1.1(a). Its Hamiltonian can be written as

$$\begin{aligned} \hat{H}_{\text{AIM}} = & \sum_{\sigma} \epsilon \hat{f}_{\sigma}^{\dagger} \hat{f}_{\sigma} + U \hat{f}_{\uparrow}^{\dagger} \hat{f}_{\downarrow}^{\dagger} \hat{f}_{\downarrow} \hat{f}_{\uparrow} \\ & + \sum_{k,\sigma} \epsilon_k \hat{c}_{k\sigma}^{\dagger} \hat{c}_{k\sigma} + \sum_{k,\sigma} \left(V_k \hat{f}_{\sigma}^{\dagger} \hat{c}_{k\sigma} + V_k^* \hat{c}_{k\sigma}^{\dagger} \hat{f}_{\sigma} \right), \end{aligned} \quad (1.1)$$

with spin $\sigma \in \{\uparrow, \downarrow\}$, impurity creation (annihilation) operators $\hat{f}^{\dagger}$ ($\hat{f}$), and bath creation (annihilation) operators $\hat{c}^{\dagger}$ ($\hat{c}$). The first two terms of Eq. (1.1) can be identified as the impurity Hamiltonian $\hat{H}_i$.

According to Lu et al. [15], we can use different bath geometries related to each other by unitary transformations, as we want to represent the ground state of the system with the fewest Slater determinants (SDs) possible. Since it is not convenient to mix impurity and

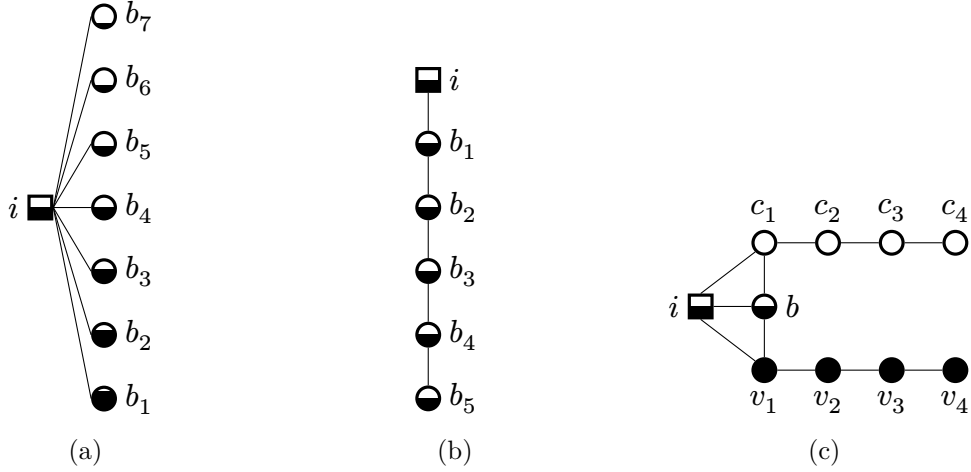


Figure 1.1: Different bath geometries. Impurity i interacting with bath sites b_i , or b , v_i , c_i . Site occupation is indicated by fill level. (adapted from [15], Fig. 1)

mean-field approximate sites, only basis rotations within the mean-field sites are allowed, preserving $\hat{H}_i$.

The advantage of the representation in Fig. 1.1(a) is that bath sites with low (high) onsite energy are basically full (empty), indicated by their fill level. Hopping is annotated by connecting sites by lines. The disadvantage is that each bath site can interact with the impurity and therefore is important.

We can make a basis transformation in which the impurity interacts with only one bath site, which in turn interacts with the next bath site etc., forming a chain. This is shown in Fig. 1.1(b) where the single particle Hamiltonian can be expressed in a tridiagonal form. The problem in this representation is that each bath site is partially occupied, and the ground state requires an exponentially growing number of SDs.

The minimal number of SDs to describe the ground state can be achieved in the natural orbital basis. In this representation we create full (valence bath sites v_i) and empty chains (conduction bath sites c_i) which are coupled to the impurity and a mirror bath site b as depicted in Fig. 1.1(c). In order to be able to have an arbitrary filling n of the impurity, the mirror bath site has the opposite filling $1 - n$ [15].

We can further split the setup into two components as shown in Fig. 1.2. The first component only includes the impurity, mirror bath site, and L valence and conduction bath sites, with L as a tunable parameter [12]. In general this value can differ for conduction/valence bath sites. The second component includes all other bath sites, which we want to approximate with CI. This splits the Hamiltonian into three components (see Fig. 1.2): $\hat{H}_s$ which acts on the impurity and sites close to it, $\hat{H}_v$ which acts on the truncated chains, and $\hat{H}_{\text{mix}}$ which connects both. For no excitation inside the truncated chains, all valence bath sites v_i are filled and all conduction bath sites c_i are empty which we write as $|\mathbb{1}_v\rangle \otimes |\mathbb{0}_c\rangle$ [12]. In this work we will limit the number of excitations to singles and doubles, meaning that up to one (two) occupations of spin-sites can differ from the aforementioned state.

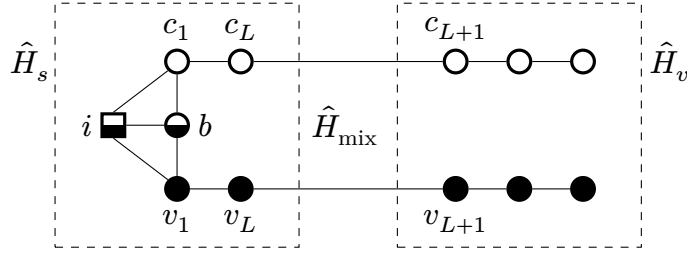


Figure 1.2: Separating the Anderson Hamiltonian $\hat{H}_{\text{AIM}}$ into three components: $\hat{H}_s$, $\hat{H}_v$ act on sites enclosed by its corresponding box, $\hat{H}_{\text{mix}}$ connects the two parts. (adapted from [12], Fig. 2)

1.2 Fermions package

We implement the concepts and ideas described in the previous section in the module `WAVEFUNCTIONS` of the `FERMIONS` [16] package, which is written in Julia [17]. First, we will give a short introduction of the package and status quo, which uses the occupation number basis for SDs. Each site k of the AIM contains two spin-sites σ with occupation $n_{k\sigma} \in \{0, 1\}$ for fermions. Each SD $|n_n, \dots, n_3, n_2, n_1\rangle$ can therefore be interpreted as a bit pattern of zeros and ones represented by an unsigned integer with creation/annihilation operators acting as bit flips. The vacuum state $|0\rangle$ can be depicted as zero consisting of no particles. To store the wave function, we can use a dictionary with keys acting as SDs and associated values acting as probability amplitudes

$$|\psi\rangle = \sum_i \alpha_i |S_i\rangle, \quad (1.2)$$

which is currently implemented in the struct `Wavefunction`. For normalization $\sum_i |\alpha_i|^2 = 1$ must be fulfilled. To satisfy the fermionic anticommutator relations, the annihilation operator is defined as

$$\begin{aligned} \hat{c}_k |\dots, n_k = 1, \dots\rangle &:= \prod_{i=1}^{k-1} (-1)^{n_i} |\dots, n_k = 0, \dots\rangle \\ &= (-1)^{\sum_{i=1}^{k-1} n_i} |\dots, n_k = 0, \dots\rangle. \end{aligned} \quad (1.3)$$

2 Implementation

Currently, the limiting factor is the lookup of bit patterns inside the dictionary. We therefore want to reduce the number of dictionary keys, which we implement under the name “hybrid representation” (HR).

2.1 Hybrid wave function

For the geometry shown in Fig. 1.2 we can define two sectors. The first contains all sites on which $\hat{H}_s$ acts, the second contains the sites in the two chains of $\hat{H}_v$. Given a fixed number of electrons we can distribute them over these sectors such that for each sector we obtain a set of electronic configurations. Every single determinant state can be expressed by combining one configuration from each sector: $|s_i\rangle$ from $\hat{H}_s$ and $|v_i\rangle$ from $\hat{H}_v$. The first sector uses the existing bit representation (BR) to denote a configuration. In the second sector however, we map the configurations onto linearly independent vectors $|v_i\rangle = \sum_{\alpha} v_{i\alpha} |\alpha\rangle$ with each index α corresponding to a given state $|\alpha\rangle$ of the truncated chains and the value $v_{i\alpha}$ corresponding to the probability amplitude. This way every state of the full system can be expressed as

$$|\psi\rangle = \sum_i |s_i\rangle \otimes |v_i\rangle = \sum_{i,\alpha} v_{i\alpha} |s_i\rangle \otimes |\alpha\rangle. \quad (2.1)$$

Normalization is fulfilled by

$$\sum_i \| |v_i\rangle \|^2 = \sum_{i,\alpha} |v_{i\alpha}|^2 = 1. \quad (2.2)$$

For a system depicted in Fig. 1.2 limited to singles, and starting with 3 empty and 3 filled states in $\hat{H}_v$, and a factor 2 for spin, the vector dimension would be $1 + 2 \cdot (3 + 3) = 13$; one entry stores the amplitude for no excitation and 12 entries for an excitation in any of the possible spin-sites.

For this, we can use the struct `CIWavefunction`.

```
struct CIWavefunction{T,nslater,nfilled,nempty,excitation}
    terms::T
end
```

Determinants with their corresponding vectors are stored inside `terms` as a dictionary. To uniquely describe this new wave function, we must save the number of sites inside the determinant n_s , number of filled (empty) sites n_f (n_e) inside the vector, and maximum possible CI excitations e . In this way, the structure is defined at compile time rather than runtime if these values were field names instead. We can define the interface as:

- `terms()` returning the dictionary,
- `nslater()` returning n_s ,
- `nfilled()` returning n_f ,
- `nempty()` returning n_e ,
- `excitation()` returning e ,
- `nvector()` returning $n_v = n_f + n_e$,
- `nsites()` returning $n_s + n_f + n_e$.

2.1.1 Vector indexing

For vector indexing we must establish a bijective mapping between indices and states. We could group the spin-sites inside the SD either by spin

$$\underbrace{|\dots, c_{2\uparrow}, c_{1\uparrow}, \dots, v_{2\uparrow}, v_{1\uparrow}, s_{\uparrow}, \dots, c_{2\downarrow}, c_{1\downarrow}, \dots, v_{2\downarrow}, v_{1\downarrow}, s_{\downarrow}\rangle}_{\uparrow}, \quad (2.3)$$

or by site

$$|\dots, \underbrace{c_{2\uparrow}, c_{2\downarrow}}_{c_2}, \underbrace{c_{1\uparrow}, c_{1\downarrow}}_{c_1}, \dots, \underbrace{v_{2\uparrow}, v_{2\downarrow}}_{v_2}, \underbrace{v_{1\uparrow}, v_{1\downarrow}}_{v_1}, s\rangle. \quad (2.4)$$

To decide between these two possibilities, we can calculate the effect of a hopping term $t_{ij}\hat{c}_i^\dagger\hat{c}_j$ on a SD using the anticommutator definition given in Eq. (1.3). For $i > j$ we get

$$\begin{aligned} t_{ij}\hat{c}_i^\dagger\hat{c}_j|\dots, n_i = 0, \dots, n_j = 1, \dots\rangle \\ = t_{ij} \prod_{k=j+1}^{i-1} (-1)^{n_k} |\dots, n_i = 1, \dots, n_j = 0, \dots\rangle, \end{aligned} \quad (2.5)$$

and for $i < j$

$$\begin{aligned} t_{ij}\hat{c}_i^\dagger\hat{c}_j|\dots, n_j = 1, \dots, n_i = 0, \dots\rangle \\ = t_{ij} \prod_{k=i+1}^{j-1} (-1)^{n_k} |\dots, n_j = 0, \dots, n_i = 1, \dots\rangle, \end{aligned} \quad (2.6)$$

As we can see, the sign of the result depends on the number of occupied sites between i and j . Since our chains only contain nearest-neighbor hopping, we can group by spin and skip the sign calculation.

Let the superscript denote the excitation and the subscript the index within the vector. We can store excitations sequentially, e.g., if $|v^0\rangle = (v_1^0)$ contains no excitations, $|v^1\rangle = (v_1^1, v_2^1, \dots)$ all single excitations, etc., the full vector can then be assembled by listing the entries of each component $|v\rangle = (v_1^0, v_1^1, v_2^1, \dots)$. The vector dimension is therefore

$$\dim(|v\rangle) = \sum_{i=0}^e \dim(|v^i\rangle) = \sum_{i=0}^e \binom{2n_v}{e}, \quad (2.7)$$

where n_v is the number of sites inside $\hat{H}_v$ and the factor 2 accounts for spin. Since $\dim(|v^0\rangle) = 1$, no index calculations are necessary for no excitation. For single excitation inside $|v^1\rangle$, we must establish a mapping

$$f : \{1, \dots, n_v\} \times \{1, 2\} \rightarrow \{1, \dots, 2n_v\}, \quad (2.8)$$

by assigning each spin-site $(k, \sigma) \mapsto k + (\sigma - 1)n_v =: a$ an index a and storing the amplitude of this excitation in v_a^1 . This function f and its inverse are implemented under `_rank1()` and `_unrank1()`.

For higher excitations we have multiple values a_i , which we again want to map to an index. For this, we can use the combinatorial number system (combinadics) [18]. Given a sequence of numbers $2n_v \geq a_k > \dots > a_2 > a_1 \geq 0$, it assigns a zero-based index N by the sum of binomial coefficients

$$N = \sum_{i=1}^k \binom{a_i}{i}, \quad (2.9)$$

with $\binom{n}{k} = 0$ for $k > n$. This operation is called “ranking”, providing a one-to-one mapping between sequence of numbers and integers without any gaps [19]. The inverse operation (“unranking”) can be done iteratively. In the first step we find the biggest integer a_k which satisfies

$$a_k = \max_{a_k \in \mathbb{N}} \left(\binom{a_k}{k} \leq N \right). \quad (2.10)$$

The next number can be found by subtracting the result from the previous index.

$$a_{k-1} = \max_{a_{k-1} \in \mathbb{N}} \left(\binom{a_{k-1}}{k-1} \leq N' \right), \quad N' = N - \binom{a_k}{k}. \quad (2.11)$$

For double excitation however, we can use analytical expressions

$$a_2 = \left\lfloor \frac{1 + \sqrt{1 + 8N}}{2} \right\rfloor, \quad (2.12)$$

$$a_1 = N - \binom{a_2}{2}. \quad (2.13)$$

Since Julia indexing is one-based, numbers must be shifted accordingly. Mapping between spin-sites and index are available under the functions `getvectorindex()` returning the vector index given excited spin-sites, and `getspinsites()` returning excited spin-sites given a vector index.

2.2 Hybrid operator

To represent the Anderson Hamiltonian which acts on a hybrid wave function, we can create a struct `CI0operator`. As mentioned in Section 1.1 it should contain three parts,

one operator acting on $|s\rangle$, one acting on $|v\rangle$ using an expansion, and a mixed operator representing hopping between $|s\rangle$ and $|v\rangle$

$$\hat{H} = \hat{H}_s + \hat{H}_v + \hat{H}_{\text{mix}}. \quad (2.14)$$

The idea is to implement $\hat{H}|\psi\rangle$ by applying these Hamiltonians successively. For $\hat{H}_s|s\rangle$ we can use the existing FERMIONS code with minimal changes. This is implemented under the function `_slaterop!()`.

The next term, $\hat{H}_v$ can be written as

$$\hat{H}_v = \sum_{k,\sigma} \epsilon_k \hat{c}_{k\sigma}^\dagger \hat{c}_{k\sigma} - \sum_{\langle ij \rangle, \sigma} (t_{ij} \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} + t_{ij}^* \hat{c}_{j\sigma}^\dagger \hat{c}_{i\sigma}), \quad (2.15)$$

with energy levels ϵ_k and hopping amplitudes t_{ij} . $\langle \rangle$ denotes nearest-neighbor hopping. The expansion by excitation is given by

$$\hat{H}_v \approx \hat{H}_{v0} + \hat{H}_{v1} + \hat{H}_{v2} + \dots = \sum_{i=0}^e \hat{H}_{vi}. \quad (2.16)$$

In the following we will explain each term of the expansion and how to store and apply it to a hybrid wave function efficiently. As the Hamiltonian in Eq. (2.15) conserves the particle number

$$[\hat{H}_v, \hat{N}] = \sum_{k,\sigma} [\hat{H}_v, \hat{c}_{k\sigma}^\dagger \hat{c}_{k\sigma}] = 0, \quad (2.17)$$

we can treat each excitation separately. For the struct it is enough to store the energy levels and hopping amplitudes inside vectors (ϵ , $\mathbf{t}$ respectively), since mapping between indices and sites is established in Section 2.1.1. In the current implementation we could either supply vectors with $\dim(\epsilon) = n_v$ and $\dim(\mathbf{t}) = n_v - 1$ internally copying them for both spin orientations, or $\dim(\epsilon) = 2n_v$ and $\dim(\mathbf{t}) = 2(n_v - 1)$ which sets spin-dependent energy and hopping ($\epsilon_k \rightarrow \epsilon_{k\sigma}$ and $t_{ij} \rightarrow t_{ij\sigma}$). At the moment both vectors are restricted to real values.

Based on the information of ϵ and $\mathbf{t}$ it is possible to get the result of any excitation Hamiltonian $\hat{H}_{ve}$ applied on the vector $|v^e\rangle$ without creating the matrix explicitly (matrix-free method). The disadvantage would be that each matrix-vector result must be calculated on-the-fly, e.g., for a hybrid representation limited to double excitation (HR-2) with 100 determinants, a total of $3 \cdot 100$ matrix-vector calculations must be done with the factor 3 resulting from the maximum excitation (including no excitation). As we want this to be fast, this is not ideal. We could therefore create a matrix for each excitation once and cache it if it is feasible memory-wise. They can then be applied onto each vector using built in matrix-vector multiplication which has been optimized for decades already. In the aforementioned example three matrices would be created.

For no excitation, the hopping terms of $\hat{H}_v$ in Eq. (2.15) can be ignored since all bath sites are either full (v_k) or empty (c_k). The Hamiltonian is therefore just a scalar

$$\hat{H}_{v0} = \sum_{v_k, \sigma} \epsilon_{v_k} =: \epsilon_{v0}. \quad (2.18)$$

Given an excitation e , we should first estimate the number of nonzero entries N_{nz} inside $\hat{H}_{ve}$ to determine whether we could even store the matrix. If an excitation is at location (k, σ) , we must store one value on the diagonal for the energy $\epsilon_0 \mp \epsilon_{k\sigma}$ with the sign depending on whether a valence or conduction bath site is excited, and two values for nearest-neighbor hopping $(k, \sigma) \rightarrow (k \pm 1, \sigma)$. This results altogether in

$$N_{\text{nz}} < (1 + 2e) \binom{2n_v}{e} \quad (2.19)$$

with the inequality resulting from the fact that excitations in the first and last spin-site can only hop in one direction. Using this result, we can estimate the fill factor of a matrix, which is defined as the ratio of nonzeros to matrix dimension

$$f < \frac{(1 + 2e) \binom{2n_v}{e}}{\binom{2n_v}{e}^2} \ll 1. \quad (2.20)$$

Since we assume that $e \ll n_v$, this value is much smaller than one, which suggests a sparse matrix storage method. The Julia documentation¹ lists the compressed sparse column (CSC) format. We can set an upper bound of the storage size using the coordinate format (COO) keeping three 64 bit numbers for each nonzero entry. As displayed in Table 2.1, this poses no problem for singles and doubles, while triples could be problematic for consumer grade hardware.

Table 2.1: Storage estimation (upper bound) in byte of the Hamiltonian in COO format.

Excitation	n_b (bath sites)		
	100	200	300
Single	14×10^3	29×10^3	43×10^3
Double	2×10^6	10×10^6	22×10^6
Triple	221×10^6	2×10^9	6×10^9

For single excitation we can use a more optimized storage format. As the matrix is tridiagonal, we can use the existing structs `Tridiagonal` for complex values or `SymTridiagonal` for real values.

Currently, the modified FERMIONS package features up to double excitation. Matrices are cached when creating a new instance of `CIOperator`. The matrix-vector multiplication between $\hat{H}_v$ and $|v\rangle$ is implemented inside the function `_vectorop!()` using five-argument `mul!()`.

Lastly, let us turn to $\hat{H}_{\text{mix}}$. We can understand the mixed operator as removing a particle from a spin-site $n_{i\sigma}$ inside $|s\rangle$, and adding it to spin-site $n_{j\sigma}$ inside $|v\rangle$ or vice

¹<https://docs.julialang.org/en/v1/stdlib/SparseArrays/>

versa. This can be stored inside the struct `CIOperatorMixed`. Constructing this object works from tuples (i, j, t) which automatically creates four operators

$$t\hat{c}_{i\uparrow}^\dagger\hat{c}_{j\uparrow}, \quad t\hat{c}_{j\uparrow}^\dagger\hat{c}_{i\uparrow}, \quad t\hat{c}_{i\downarrow}^\dagger\hat{c}_{j\downarrow}, \quad t\hat{c}_{j\downarrow}^\dagger\hat{c}_{i\downarrow}. \quad (2.21)$$

We can calculate $\hat{H}_{\text{mix}}|\psi\rangle$ by first applying the corresponding operator on $|s\rangle$ using already existing infrastructure. As the sign of the phase depends on the occupation of all spin-sites between i and j (see Eqs. (2.5) and (2.6)), we can split it into two components: a bit phase and vector phase. For the former we can store a mask which considers all sites following (i, σ) , e.g., if it is at the fourth location, the mask contains ones from digit five onwards (...11110000). We can interpret adding/removing a particle in $|v\rangle$ as lowering or raising the excitation and store it as a boolean. If the excitation inside $|v\rangle$ is lowered, the single excitation state on (j, σ) is mapped to the state without excitation. The excitation $c_{j\sigma}^{(\dagger)}$ must be at the first valence (conduction) bath site. If the excitation j is at the first valence bath site, the vector phase is one; otherwise the excitation is at the first conduction bath site, resulting in the sign $(-1)^{n_f}$. For double excitation at $((j, \sigma), (k, \sigma'))$, the state is mapped to the single excitation state (k, σ') . If (k, σ') has a smaller index than (j, σ) , the aforementioned phase is multiplied by (-1) , since one particle more/less exists. Raising the excitation can be done similarly.

3 Results

To characterize the implementation of the hybrid representation, we can create a single-determinant state $|\psi_0\rangle$ and do a power iteration to generate new states containing multiple determinants

$$|\psi_n\rangle = \frac{\hat{H} |\psi_{n-1}\rangle}{\|\hat{H} |\psi_{n-1}\rangle\|} = \frac{\hat{H}^n |\psi_0\rangle}{\|\hat{H}^n |\psi_0\rangle\|}. \quad (3.1)$$

This will be relevant when calculating the ground state using the Lanczos algorithm [20] which uses a Krylov subspace

$$\mathcal{K}_n(\hat{H}, |\psi\rangle) = \text{span}\{|\psi\rangle, \hat{H} |\psi\rangle, \hat{H}^2 |\psi\rangle, \dots, \hat{H}^n |\psi\rangle\} \quad (3.2)$$

for a given operator $\hat{H}$ and state $|\psi\rangle$. We create a Hamiltonian for the geometry shown in Fig. 1.1(b) using the parameters listed in Table 3.1. This Hamiltonian is converted to the natural orbital representation and the starting state is half-filled as shown in Fig. 1.1(c)

$$|\psi_0\rangle = f_{\uparrow}^{\dagger} \hat{c}_{b\downarrow}^{\dagger} \prod_{v_k, \sigma} \hat{c}_{v_k \sigma}^{\dagger} |0\rangle. \quad (3.3)$$

$\hat{c}_b^{\dagger}$ is the creation operator for the mirror bath site.

Performance benchmarks ran on the Vienna Scientific Cluster VSC-5 using the package BENCHMARKTOOLS [21]. The minimum time was used as the runtime estimator following the authors' suggestion. The system contains AMD EPYC™ 7713¹ CPUs with 64 cores and at least 256 GB of memory. As the implemented code is currently single-threaded and memory requirements depend on the model complexity, it can also run on consumer hardware.

Table 3.1: Hamiltonian parameters used for benchmarks.

Parameter	Symbol	Value
Coulomb interaction	U	4.0
Impurity energy level	ϵ	-2.0
Bath energy levels	ϵ_k	0.0
Hybridization	V	-1.0
Bath hopping	$-t_{ij}$	-1.0

¹<https://www.amd.com/en/products/cpu/amd-epyc-7713>

3.1 Numerical precision

In order to compare results between the bit and hybrid representation (BR and HR respectively), we can write functions to convert between them. We calculate up to 20 iterations, varying the amount of bath sites, and number of sites inside the determinant of the HR. After a couple of iterations we can find a mismatch of SDs, which are associated with amplitudes $|\alpha_i| \leq 10^{-20}$ which are much smaller than machine epsilon $\epsilon = 2^{-52} \approx 10^{-16}$, and generated when two amplitudes add up destructively. After removing all SDs with weight $|\alpha_i| < \epsilon$, each amplitude obtained by the two methods can be compared against each other and verified to fit within a relative tolerance of 10^{-10} . For the whole wave function the difference is roughly

$$\| |\psi^{\text{BR}} \rangle - |\psi^{\text{HR}} \rangle \| \approx O(10^{-15}). \quad (3.4)$$

3.2 Hamiltonian

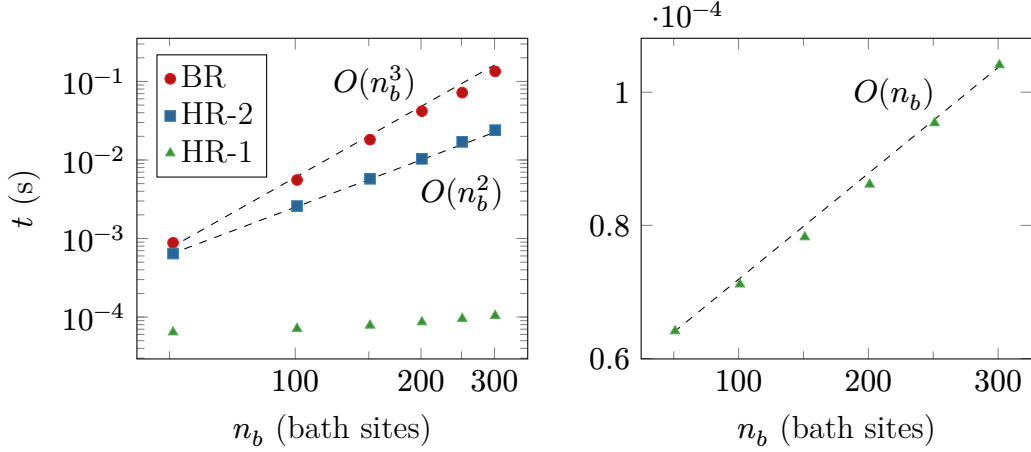


Figure 3.1: Time taken to create the Hamiltonian in the natural orbital representation using different methods. The hybrid representation (HR) uses four sites (including impurity) in the bit component, single (HR-1) and single plus double (HR-2) excitations in $\hat{H}_v$. Right panel shows HR-1 using linear axes.

For the following tests we put four sites (including impurity) into the bit component and all other sites into the vector component. Figure 3.1 shows the time taken to create the Hamiltonian for both representations. The old method shows a cubic scaling while the new method scales by the maximum allowed excitation.

3.3 Power iteration

Since normalization is faster than applying the Hamiltonian, we only time the numerator of Eq. (3.1) in benchmarks. The top panels of Fig. 3.2 show the time taken to calculate

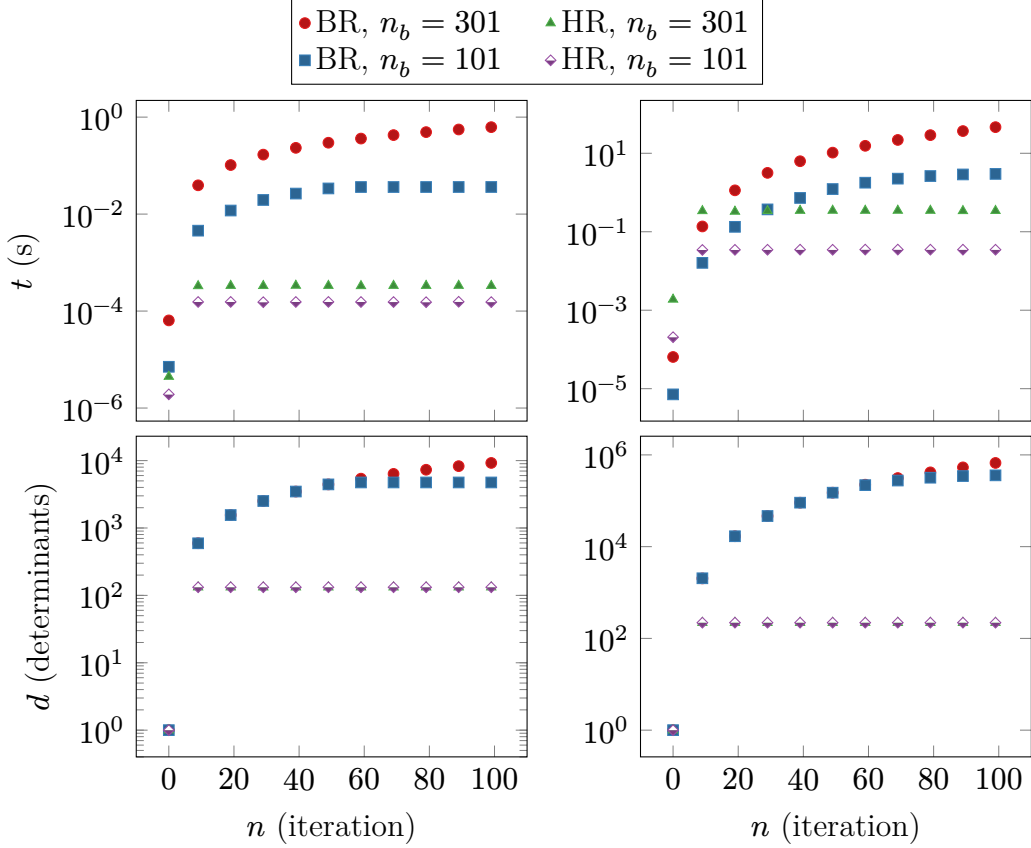


Figure 3.2: Power iteration n of Eq. (3.1) using 101 and 301 bath sites n_b . Top panels show time to calculate one step $\hat{H}|\psi_n\rangle$ excluding normalization. Bottom panels show number of determinants of $|\psi_n\rangle$, which overlap for the hybrid representation. Left: single excitation. Right: single plus double excitation.

one iteration $\hat{H}|\psi_n\rangle$. We can see that the time for the HR plateaus already at iteration ten. This flattening can be understood if we look at the number of determinants at each step, shown in the bottom panels of Fig. 3.2. Since most of the information is stored inside vectors and not the bit component (containing only four sites), the number of possible determinants is very low and independent of the number of bath sites. For a half-filled system we get 36 determinants in a state of no excitation, 96 in a single excitation state, and 88 in a double excitation state. Once these determinants are created, which happens before the tenth iteration, there is no additional cost, and therefore constant time.

The BR with 101 bath sites restricted to singles plateaus between iteration 50 and 60. This happens because we have two chains with 50 bath sites each. The excitation starts at the beginning of the chain and hops to the next site at each iteration. Once we reach the end, all possible states are generated.

To get the improvement of the new method over the old one, we can create the ratio of the measured times which is shown in Fig. 3.3. We can see that the new method restricted

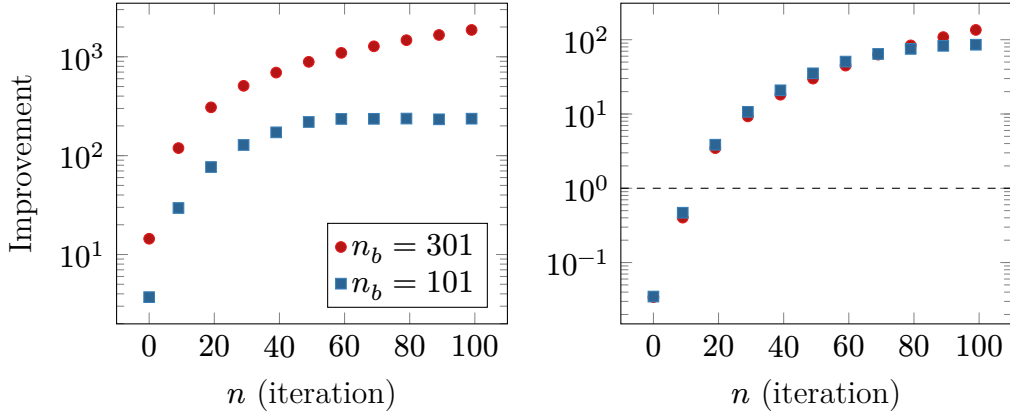


Figure 3.3: Performance improvement of the new HR over the existing BR when calculating one step $\hat{H}|\psi_n\rangle$ of the power iteration excluding normalization. HR uses four sites (including impurity) in the bit component. Here, 101 and 301 bath sites n_b were tested. Left: single excitation. Right: single plus double excitation.

to singles shows an improvement for all measured parameters. In the case of 101 bath sites the improvement plateaus between iteration 50 and 60 which we explained before.

For double excitation we see a performance regression for the first 15 iterations which correspond to states with less than around 10^4 SDs. For states with more determinants we gain a performance boost with the HR.

We can take the measured value at the highest iteration which corresponds to the calculation of $\hat{H}|\psi_{99}\rangle$ and compare it against the number of bath sites n_b , as shown in Fig. 3.4. HR-1 shows a good agreement with a linear fit, while for HR-2 we can see a scaling which is a little worse than $O(n_b^2)$.

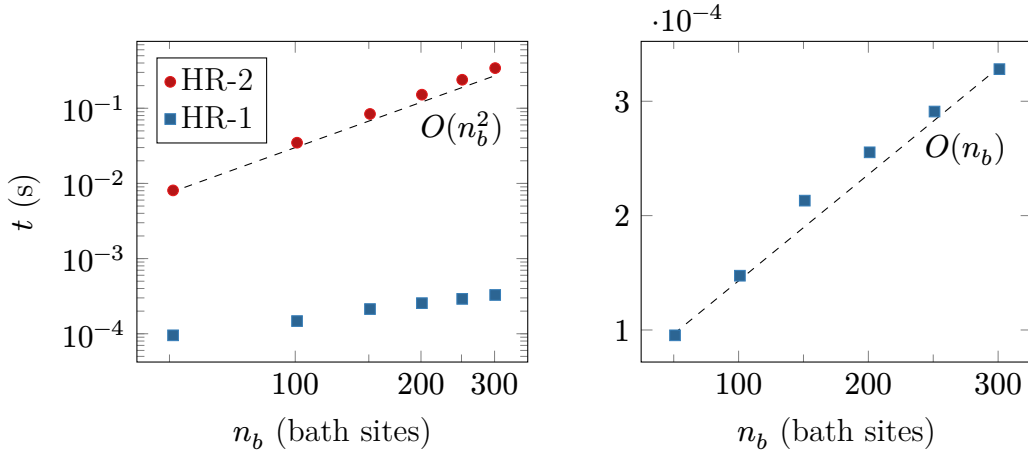


Figure 3.4: Time to calculate $\hat{H}|\psi_{99}\rangle$ in the hybrid representation for singles (HR-1) and singles plus doubles (HR-2).

4 Conclusions and outlook

We have shown that by splitting the AIM into two components, one part can be represented as bits, and another can be represented as a vector of CI excitations. This allows us to speed up the calculation $\hat{H}|\psi\rangle$ of the Anderson Hamiltonian on a state by up to a factor of 2000 depending on the number of bath sites and excitation with an accuracy loss of $O(10^{-15})$ for the wave function. Calculations which took tens of seconds can now be done in a matter of hundreds of ms.

Using this new tool, our goal is to determine properties and Green's functions of systems which contain hundreds of bath sites which is going to be the work of my master thesis.

Although the current implementation shows numerical consistence with the existing bit representation, some restrictions remain. One is that hopping is currently restricted to real values, which is not hard to extend to the complex domain if necessary. Another factor is the addition of a spin dependency to the mixed operator $\hat{H}_{\text{mix}}$. Instead of using tuples (i, j, t) , we can use (i, j, σ, t) and adjust the backend which is not much work. If this feature is realized, all three terms of Eq. (2.14) can accommodate spin-dependent energies and hopping.

We can extend the feature set to triple excitations if doubles are not accurate enough. For fast unranking (see Eq. (2.10)) this requires solving a cubic equation on integers and storage of a matrix which would take a few GB of space (Table 2.1).

At the moment, terms of opposite spin hopping, e.g., $\hat{c}_{i\uparrow}^\dagger \hat{c}_{j\downarrow}$ will evaluate with a possible wrong sign in the bit component. These terms are not necessary as long as we have time reversal symmetry, since we can change into the $\hat{J}$, $\hat{J}_z$ basis. If one wants to implement a few of this kind of terms, it is possible to write a custom operator, otherwise it would probably be better to change the whole underlying order from spin-sorting defined in Eq. (2.3) to site-sorting (Eq. (2.4)), which requires a major rewrite.

To get even more performance, we could employ multithreading. The first choice would be the application of the vector Hamiltonian on the vector component $\hat{H}_v|v\rangle$, since it does not change dictionary keys and each matrix-vector multiplication is independent of each other.

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